MSDI Calculator

Manure Application Equipment Performance Methodology

Technical Methodology Memorandum

Prepared for internal model documentation and technical review

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Model scope: Manure Sub-Surface Drip Irrigation (mSDI-e) economic screening calculator

Document status: MSDI calculator app logic and methodology surrounding in-field manure application.

Purpose and Scope

This technical memorandum documents the screening-level manure application equipment performance coefficients used by the MSDI Calculator. The lookup values are used to estimate application labor time, fuel use, electricity use, and operational cost for manure land-application pathways. The focus is on two coefficient classes: application rate, expressed as wet tons per hour, and fuel use rate, expressed as gallons of diesel per hour. Electricity use rates are retained as lookup inputs where the pathway is represented as electrically pumped irrigation rather than diesel-powered field equipment.

Relationship to the MSDI Calculator

The manure application lookup table supports operational-expense calculations for existing and proposed manure handling systems. The model converts manure quantity into application time using an equipment-specific application rate. The resulting hours are then multiplied by fuel, electricity, labor, or other cost factors elsewhere in the calculator. Because the lookup values influence annual manure application costs, they should represent realistic screening-level equipment performance rather than exact site-specific capacity.

Terminology and Units

The lookup table uses wet tons per hour for application rate. For liquid manure pathways, wet tons per hour is converted from volumetric flow using a density assumption of 8.5 pounds per gallon and 2,000 pounds per ton. Fuel use is expressed as gallons of diesel per operating hour. Electricity use is expressed as kilowatt-hours per operating hour where the pathway represents an electrically powered irrigation or pumping system.

Selected Screening-Level Coefficients

Table 1 summarizes the updated manure application lookup values. These values are intended to be representative coefficients for economic screening. They are not intended to replace farm-specific calibration, custom applicator invoices, flow-meter records, or field logs.

Manure Application Pathway	Application Rate	Fuel Rate	Electricity Rate	Implied Liquid Flow	Basis
Liquid-Drag Line	250 tons/hr	12 gal/hr	0 kWh/hr	≈58,800 gal/hr; ≈980 gpm	Lower-end continuous draghose / dragline flow basis.
Liquid-Honey Wagon	200 tons/hr	12 gal/hr	0 kWh/hr	≈47,100 gal/hr; ≈785 gpm	Updated from 38 tons/hr; discharge-rate basis rather than full haul-cycle capacity.
Liquid-HP Irrigation (Pivot or Gun)	250 tons/hr	0 gal/hr	75 kWh/hr	≈58,800 gal/hr; ≈980 gpm	Electric pumping row; diesel fuel not assigned.
Liquid-LP Irrigation (Furrow or drip)	125 tons/hr	0 gal/hr	8 kWh/hr	≈29,400 gal/hr; ≈490 gpm	Lower-pressure liquid distribution row; diesel fuel not assigned.
Solid-Surface Spreading	40 tons/hr	7 gal/hr	0 kWh/hr	N/A	Screening-level solid spreading capacity with moderate tractor load.
Solid-Injected (Strip Till)	30 tons/hr	12 gal/hr	0 kWh/hr	N/A	Lower throughput than surface spreading due to injection draft and field-speed constraints.
N/A	0 tons/hr	0 gal/hr	0 kWh/hr	N/A	No application pathway selected.

Core Conversion Methodology

For liquid manure pathways, the lookup table stores application rate in wet tons per hour to align with the existing manure application OPEX calculation structure. Where a literature or equipment source reports flow in gallons per minute or gallons per hour, the value is converted to tons per hour using the assumed liquid manure density.

Equation 1. Tons per hour = Gallons per hour × 8.5 lb/gal ÷ 2,000 lb/ton

Equation 2. Gallons per hour = Gallons per minute × 60

Equation 3. Gallons per minute = Tons per hour × 2,000 lb/ton ÷ 8.5 lb/gal ÷ 60

Part 1. Liquid Manure Application Rates

1.1 Honey Wagon / Tank Spreader

The prior honey-wagon value of 38 tons per hour was not consistent with typical liquid-manure discharge capacity. At 8.5 lb/gal, 38 tons per hour is only about 8,940 gal/hr, or 149 gpm. Michigan State University Extension reports representative tank spreader unloading rates of 900 to 1,550 gpm, with a typical unloading rate of about 1,300 gpm for surface spreading. The updated screening value of 200 tons per hour corresponds to approximately 47,100 gal/hr, or about 785 gpm. This is below the reported MSU unloading-rate range and therefore conservative when interpreted as discharge/application capacity rather than total haul-cycle capacity.

Equation 4. 200 tons/hr × 2,000 lb/ton ÷ 8.5 lb/gal = 47,059 gal/hr

Equation 5. 47,059 gal/hr ÷ 60 = 784 gal/min

The selected value intentionally does not attempt to model hauling distance, road speed, loading delay, maneuvering, field conditions, or nurse-tank logistics. MSU emphasizes that the full manure-hauling cycle includes load, transport, unload, and return, and that travel distance can materially change achieved hauling rate. The MSDI Calculator uses the lookup coefficient as a screening-level application equipment rate rather than a complete haul-cycle simulation.

1.2 Dragline / Draghose

The dragline pathway is represented as continuous or near-continuous liquid manure transfer and field application. A Jefferson County draghose fact sheet reports typical draghose flow rates of approximately 400 to 1,000 gpm, depending on pump capacity, manure flowability, pipe size, pipe length, and elevation. The selected value of 250 tons per hour implies approximately 980 gpm, placing the lookup value near the upper end of that reported typical range. This is appropriate for a screening-level dragline coefficient because dragline systems avoid repeated tank hauling cycles and are often selected for high-throughput application.

Equation 6. 250 tons/hr × 2,000 lb/ton ÷ 8.5 lb/gal ÷ 60 = 980 gal/min

1.3 High-Pressure and Low-Pressure Irrigation Pathways

The high-pressure irrigation pathway is retained at 250 tons per hour, equivalent to approximately 980 gpm. The low-pressure irrigation pathway is retained at 125 tons per hour, equivalent to approximately 490 gpm. These rows are represented as electrically powered pathways in the lookup table, so diesel fuel use remains zero and electricity-use assumptions are carried separately. Manufacturer and applicator guidance for liquid manure systems commonly expresses field application as a function of flow rate, target gallons per acre, toolbar or wetted width, and ground speed; therefore, the selected values should be interpreted as hydraulic throughput coefficients rather than fixed agronomic application rates.

Equation 7. 125 tons/hr × 2,000 lb/ton ÷ 8.5 lb/gal ÷ 60 = 490 gal/min

Part 2. Solid Manure Application Rates

Solid manure application is more variable than liquid manure application because capacity depends on manure density, bedding content, load size, spreader geometry, travel distance, spread width, field speed, and target tons per acre. Purdue Extension notes that box-spreader capacity can be difficult to estimate because solid manure density varies with bedding and moisture content. Washington Dairy Nutrient Management guidance similarly emphasizes calibration based on amount applied and area covered. For this reason, the selected solid-manure coefficients are screening values rather than universal equipment ratings.

2.1 Solid Surface Spreading

The solid surface-spreading pathway is retained at 40 wet tons per hour. This represents a moderate screening-level rate for farm-scale spreading where field loading, turning, overlap, and variable density are not explicitly modeled. The value is not a substitute for spreader calibration. It is intended to provide a reasonable default for OPEX screening when site-specific tonnage-per-hour records are unavailable.

2.2 Solid Injection / Strip-Till Application

The solid-injected or strip-till pathway is retained at 30 wet tons per hour. The lower rate relative to surface spreading reflects higher draft requirements, lower operating speeds, and additional field constraints associated with incorporation or injection. The coefficient is intended to represent reduced effective field capacity rather than a change in manure density.

Fuel Use Methodology

Diesel use rates were selected using a screening-level engineering approach anchored to ASABE-style fuel consumption factors. Mississippi State University Extension states that ASABE fuel consumption factors include 0.044 gallons per horsepower-hour for diesel engines. The lookup values are rounded representative operating-hour values for the equipment class, recognizing that actual fuel use depends on engine horsepower, load factor, terrain, soil condition, transport distance, and operator behavior.

Equation 8. Diesel fuel use (gal/hr) = PTO horsepower × load factor × 0.044 gal/hp-hr

For the updated lookup table, liquid tank and dragline diesel pathways are assigned 12 gal/hr as a high-load but still screening-level value. Solid surface spreading is assigned 7 gal/hr, representing a moderate tractor load. Solid injection is assigned 12 gal/hr because injection or strip-till manure placement is expected to impose higher draft and traction demand than broadcast spreading. Electrically powered irrigation rows are assigned zero diesel fuel use because electricity is captured separately in the lookup table.

Model Use and Interpretation

The manure application lookup values are intended for comparative economic screening within the MSDI Calculator. They should be interpreted as representative equipment-performance coefficients, not as guaranteed field productivity. For individual farms, the preferred inputs would be applicator invoices, flow-meter records, tractor telematics, fuel logs, or calibrated spreader records. If site-specific data are unavailable, the lookup values provide a transparent and defensible default basis for comparing existing manure application systems against mSDI-e alternatives.

Assumptions and Limitations

- Liquid manure density is assumed to be 8.5 lb/gal for conversion between gallons and wet tons.
- Application rates represent screening-level throughput and do not explicitly include hauling distance, road transport, queueing, agitation delays, field setup, turning, or downtime.
- Honey-wagon rate is based on discharge/application capacity rather than full storage-to-field haul-cycle capacity.
- Dragline and irrigation rates are hydraulic throughput estimates and depend strongly on pump capacity, pressure, hose diameter, length, manure solids content, elevation change, and field layout.
- Solid application rates are highly sensitive to manure density, bedding content, spreader size, spread width, field speed, and target tons per acre.
- Fuel rates are rounded screening-level values based on representative horsepower/load classes, not measured fuel logs.
- Electricity-use rates for electrically pumped pathways were retained from the lookup table and were not revised in this update.

Quality-Control Checks

- Compare selected lookup values against local custom applicator invoices when available.
- Replace default rates with site-specific flow-meter or scale records where available.
- Review whether the app should distinguish discharge capacity from full haul-cycle capacity in future versions.
- Document whether manure application fuel includes only field application or also transport and agitation.
- Maintain a separate record of any future revisions to electricity-use rates for pump-based application pathways.

References and Source Links

[1] Michigan State University Extension. Manure Transport Rates and Land Application Costs: Tank Spreader. Bulletin E2767. Documents tank spreader loading and unloading rates, tank capacities, haul-cycle concepts, injection effects, and machinery cost considerations.
https://www.canr.msu.edu/resources/manure_transport_rates_and_land_application_costs_tank_spreader_e2767

[2] Mississippi State University Extension. Farm Machinery Cost Calculations. Documents ASABE-based diesel fuel consumption factor of 0.044 gal/hp-hr. https://extension.msstate.edu/publications/farm-machinery-cost-calculations

[3] Jefferson County, New York. Manure Application with a Draghose Fact Sheet. Documents typical draghose flow rates of 400 to 1,000 gpm and describes draghose system characteristics.
https://www.jeffersoncountyny.gov/media/Emergency%20Management/Agriculture%20Training/Draghose%20Fact%20Sheet

[4] Puck Enterprises. Calibrating Liquid Manure Application. Provides field formulas relating flow rate, gallons per acre, width, and ground speed, with examples using 3,000 gpm flow. https://puck.com/calibrating-liquid-manure-application/

[5] Purdue Extension. Estimating Manure Spreader Capacity. Describes capacity estimation for liquid tank, V-bottom, and box spreaders and notes variability of solid manure density. https://www.extension.purdue.edu/extmedia/AY/AY-278.html

[6] Washington Dairy Nutrient Management Program. Calibration of Manure Equipment. Defines manure application rate and provides calibration methods for solid, semi-solid, and liquid manure application.
https://www.wadairyplan.org/DNMP/calibration-of-manure-equipment

[7] University of Missouri Extension. Break-even Hauling Distance: Tractor-Pulled Manure Spreaders. Documents the importance of tractor horsepower, road speed, loads per acre, and transport distance in manure application economics. https://extension.missouri.edu/publications/g9333

Appendix A. Updated Lookup Values and Calculation Checks

Pathway	Selected rate	Implied gal/hr	Implied gpm	Check
Liquid-Drag Line	250 tons/hr	58,824	980	Within draghose typical upper range; conservative relative to large commercial systems.
Liquid-Honey Wagon	200 tons/hr	47,059	784	Below MSU tank-spreader unloading range of 900–1,550 gpm; conservative as discharge rate.
Liquid-HP Irrigation	250 tons/hr	58,824	980	Approx. 1,000 gpm hydraulic throughput.
Liquid-LP Irrigation	125 tons/hr	29,412	490	Approx. 500 gpm lower-pressure hydraulic throughput.